

Problems on String

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to find the length of a string.

Input	Hello World
Expected Output	Length = 11

2. Write a Python program to count vowels and consonants in a string.

Input	Programming
Expected Output	Vowels = 3, Consonants = 8

3. Write a Python program to check whether a string is a palindrome or not.

Input	madam
Expected Output	The string is a palindrome.

4. Write a Python program to convert a string to uppercase and lowercase.

Input	LihaTech
Expected Output	Uppercase = LIHATECH Lowercase = lihatech

5. Write a Python program to count the number of words in a sentence.

Input	Python is easy to learn
Expected Output	Word count = 5

6. Write a Python program to reverse a given string.

Input	Python
Expected Output	Reversed string = nohtyP

7. Write a Python program to check whether two strings are anagrams.

Input	listen silent
Expected Output	The strings are anagrams.

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8. Write a Python program to remove all spaces from a string.

Input	Liha Tech Kolhapur
Expected Output	LihaTechKolhapur

9. Write a Python program to replace a word in a string with another word.

Input	Sentence: I like Java Replace: Java with Python
Expected Output	I like Python

10. Write a Python program to find the frequency of each character in a string.

Input	aabcc
Expected Output	a: 2, b: 1, c: 2

11. Write a Python program to extract digits from a string.

Input	abc123xyz45
Expected Output	Digits = 12345

12. Write a Python program to check whether a string starts and ends with a given substring.

Input	String: Python Programming Start: Python End: Programming
Expected Output	Starts with Python = True Ends with Programming = True

13. Write a Python program to split a sentence into words and join them using a hyphen.

Input	Python is fun
Expected Output	Python-is-fun

14. Write a Python program to find the first non-repeating character in a string.

Input	aabbcddee
Expected Output	First non-repeating character = c

15. Write a Python program to remove duplicate characters from a string while preserving order.

Input	programming
Expected Output	Unique characters = progamin

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16. Write a Python program to count uppercase letters, lowercase letters, digits, and special characters in a string.

Input	LihaTech@2026
Expected Output	Uppercase = 2, Lowercase = 6, Digits = 4, Special = 1

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to count the frequency of each word in a sentence, ignoring letter case.

Input	Python is fun and python is powerful
Expected Output	python: 2, is: 2, fun: 1, and: 1, powerful: 1

2. Write a Python program to find the longest word in a sentence.

Input	Python makes programming enjoyable
Expected Output	Longest word = programming Length = 11

3. Write a Python program to remove punctuation marks from a string.

Input	Hello, World! Python is fun.
Expected Output	Hello World Python is fun

4. Write a Python program to find all starting positions of a substring, including overlapping occurrences.

Input	String = aaaaaa Substring = aa
Expected Output	Positions = [0, 1, 2, 3]

5. Write a Python program to rotate a string to the left by n positions.

Input	String = abcdef n = 2
Expected Output	Rotated string = cdefab

6. Write a Python program to compare two strings after ignoring spaces and letter case.

Input	String 1 = Liha Tech String 2 = lihatech
Expected Output	Strings are equal after normalization = True

7. Write a Python program to find the first and last occurrence of a given character in a string.

Input	String = programming Character = g
Expected Output	First index = 3 Last index = 10

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8. Write a Python program to compress consecutive repeated characters using character counts.

Input	aaabbcccd
Expected Output	Compressed string = a3b2c4d1

9. Write a Python program to check whether a sentence is a pangram.

Input	The quick brown fox jumps over the lazy dog
Expected Output	Pangram = True

10. Write a Python program to extract unique words from a sentence while preserving their original order.

Input	python is easy and python is powerful
Expected Output	['python', 'is', 'easy', 'and', 'powerful']

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to find the longest substring that contains no repeated characters.

Input	abcabcbb
Expected Output	Longest substring = abc Length = 3

2. Write a Python program to find the longest palindromic substring in a given string.

Input	babad
Expected Output	One longest palindromic substring = bab

3. Write a Python program to encrypt a string using a Caesar cipher with a given shift.

Input	Text = Hello World Shift = 3
Expected Output	Encrypted text = Khoor Zruog

4. Write a Python program to decode a run-length encoded string.

Input	a3b2c4
Expected Output	Decoded string = aaabbcccc

5. Write a Python program to check whether brackets in an expression are balanced.

Input	{[(())]}
Expected Output	Balanced = True

6. Write a Python program to find the longest common prefix among a list of strings.

Input	['flower', 'flow', 'flight']
Expected Output	Longest common prefix = fl

7. Write a Python program to group a list of words into anagram groups.

Input	['eat', 'tea', 'tan', 'ate', 'nat', 'bat']
Expected Output	[['eat', 'tea', 'ate'], ['tan', 'nat'], ['bat']] (group order may vary)

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8. Write a Python program to convert a snake_case identifier into camelCase.

Input	student_record_id
Expected Output	studentRecordId

9. Write a Python program to determine whether a string is made by repeating a smaller substring.

Input	abcbcabcbabc
Expected Output	Repeated pattern = abc Repeat count = 4

10. Write a Python program to find the smallest substring that contains all characters of a given pattern.

Input	String = ADOBECODEBANC Pattern = ABC
Expected Output	Minimum window = BANC